

Machine Learning Lab

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Lab 5

Part 1:

1.

```
library(data.table)
library(dplyr)
library(ggplot2)
library(slam)      # New: needed for perplexity calc
library(quanteda)  # New: to process text
library(topicmodels) # New: to fit topic models
library(word2vec)   # New: to fit word embeddings
library(tidyr)
library(purrr)
library(knitr)
library(kableExtra)
```

```
fb_congress_data <- fread("~/Documents/ML/Labs/Lab5/fb-congress-data3.csv")
dim(fb_congress_data)
```

```
[1] 6752    4
```

```
colnames(fb_congress_data)
```

```
[1] "doc_id"      "screen_name" "party"       "message"
```

The dataset contains 6752 rows and 4 columns. The variable names are `doc_id`, `screen_name`, `party`, and `message`.

```

posts_corpus <- corpus(x = fb_congress_data,
                      text_field = "message",
                      meta = list("screen_name", "party"),
                      docid_field = 'doc_id')

# print(posts_corpus)

# Tokenize & clean from particular types of words
mytokens <- tokens(x = posts_corpus,
                  remove_punct = TRUE,
                  remove_numbers = TRUE,
                  remove_symbols = TRUE,
                  remove_url = TRUE,
                  padding = FALSE)

# print(mytokens)

# Remove English stop words
mytokens <- tokens_remove(x = mytokens,
                        stopwords("en"),
                        padding = FALSE)

# Remove @'s
mytokens <- tokens_select(x = mytokens,
                        selection = 'remove',
                        valuetype = 'glob',
                        pattern = '@',
                        padding = FALSE)

# Make tokens lowercase
mytokens <- tokens_tolower(x = mytokens)

# How does it look now?
mytokens[1:3]

```

Tokens consisting of 3 documents and 2 docvars.

```

1 :
  [1] "president"      "trump"          "backs"          "paris"
  [5] "agreement"      "economic"       "environmental"  "national"
  [9] "security"       "moral"          "disaster"       "united"
[ ... and 6 more ]

```

```

2 :
[1] "many"      "thanks"    "first"     "class"     "summer"
[6] "interns"   "washington" "hard"      "work"      "folks"
[11] "#ga03"

3 :
[1] "economy"    "needs"     "shot"      "arm"       "spur"
[6] "growth"     "co-chair"  "bipartisan" "problem"   "solvers"
[11] "caucus"     "i've"
[ ... and 27 more ]

```

tokens

```

# Create document term matrix
dtm <- dfm(x = mytokens)

# Exclude words with too low frequency
dtm <- dfm_trim(dtm, min_termfreq = 5)

# Exclude documents with too low frequency
rowsums <- rowSums(dtm)
keep_ids <- which(rowsums >= 10)
dtm <- dtm[keep_ids,]

# Report final dimensionality of data set
dim(dtm)

```

```
[1] 5748 5484
```

document-term matrix 5748 (documents) 5484 (terms)

3. LDA

```

# Fit LDA (package: topicmodels)
# - 30 topics
# - Estimation algorithm: Gibbs
# - 1000 iterations
# - Set seed for replicability
# - Print iter every 1000th (I need the "quiet" model)

```

```
set.seed(1)
K <- 50
mylda <- LDA(x = dtm,
             k = K,
             method="Gibbs",
             control=list(iter = 1000,
                           seed = 1,
                           verbose = 0))
```

4.

```
# Inspect top 15 from each topics
get_terms(mylda, 15)
```

	Topic 1	Topic 2	Topic 3	Topic 4	Topic 5
[1,]	"program"	"states"	"u.s"	"need"	"week"
[2,]	"families"	"united"	"north"	"help"	"last"
[3,]	"children"	"rights"	"military"	"resources"	"night"
[4,]	"community"	"amendment"	"korea"	"crisis"	"see"
[5,]	"services"	"people"	"strategy"	"opioid"	"also"
[6,]	"need"	"right"	"administration"	"like"	"many"
[7,]	"also"	"justice"	"international"	"address"	"three"
[8,]	"grant"	"civil"	"region"	"get"	"washington"
[9,]	"seniors"	"constitution"	"world"	"epidemic"	"able"
[10,]	"centers"	"without"	"nuclear"	"drug"	"read"
[11,]	"kids"	"second"	"foreign"	"treatment"	"hear"
[12,]	"provide"	"human"	"south"	"helping"	"time"
[13,]	"start"	"individuals"	"policy"	"abuse"	"events"
[14,]	"keep"	"equality"	"syria"	"combat"	"good"
[15,]	"chip"	"americans"	"countries"	"21st"	"lot"
	Topic 6	Topic 7	Topic 8	Topic 9	Topic 10
[1,]	"state"	"president"	"americans"	"energy"	"committee"
[2,]	"today"	"trump"	"millions"	"change"	"house"
[3,]	"senator"	"administration"	"bill"	"power"	"hearing"
[4,]	"home"	"donald"	"republicans"	"climate"	"today"
[5,]	"see"	"trump's"	"million"	"clean"	"chairman"
[6,]	"west"	"trump's"	"coverage"	"future"	"member"
[7,]	"virginia"	"order"	"health"	"protect"	"subcommittee"
[8,]	"place"	"executive"	"people"	"environment"	"american"
[9,]	"across"	"j"	"plan"	"epa"	"want"

[10,]	"read"	"now"	"medicaid"	"environmental"	"oversight"
[11,]	"glad"	"decision"	"away"	"back"	"said"
[12,]	"take"	"actions"	"insurance"	"progress"	"leadership"
[13,]	"hope"	"signed"	"trumpcare"	"understand"	"well"
[14,]	"airport"	"action"	"care"	"commitment"	"including"
[15,]	"grassley"	"obama"	"conditions"	"generations"	"congresswoman"
	Topic 11	Topic 12	Topic 13	Topic 14	Topic 15
[1,]	"us"	"office"	"federal"	"life"	"general"
[2,]	"know"	"washington"	"assistance"	"history"	"investigation"
[3,]	"like"	"district"	"help"	"month"	"director"
[4,]	"fight"	"staff"	"u.s"	"story"	"russia"
[5,]	"can"	"visit"	"puerto"	"first"	"former"
[6,]	"every"	"please"	"rico"	"read"	"russian"
[7,]	"want"	"dc"	"hurricane"	"made"	"sessions"
[8,]	"many"	"learn"	"disaster"	"black"	"intelligence"
[9,]	"one"	"constituents"	"relief"	"celebrate"	"independent"
[10,]	"give"	"d.c"	"recovery"	"stories"	"fbi"
[11,]	"let"	"call"	"emergency"	"helped"	"must"
[12,]	"even"	"visiting"	"support"	"take"	"attorney"
[13,]	"share"	"tour"	"fema"	"came"	"special"
[14,]	"mr"	"capitol"	"efforts"	"shared"	"election"
[15,]	"ever"	"thank"	"provide"	"experience"	"democracy"
	Topic 16	Topic 17	Topic 18	Topic 19	Topic 20
[1,]	"education"	"law"	"system"	"great"	"honor"
[2,]	"students"	"keep"	"country"	"meeting"	"court"
[3,]	"programs"	"safe"	"immigration"	"issues"	"war"
[4,]	"science"	"enforcement"	"daca"	"yesterday"	"honored"
[5,]	"college"	"police"	"congress"	"thank"	"service"
[6,]	"workforce"	"department"	"legal"	"discuss"	"thank"
[7,]	"schools"	"communities"	"dreamers"	"thanks"	"judge"
[8,]	"young"	"protect"	"immigrants"	"meet"	"award"
[9,]	"training"	"officers"	"policy"	"met"	"supreme"
[10,]	"future"	"local"	"laws"	"association"	"ceremony"
[11,]	"career"	"safety"	"dream"	"importance"	"gorsuch"
[12,]	"technology"	"state"	"dangerous"	"leaders"	"world"
[13,]	"help"	"line"	"policies"	"enjoyed"	"justice"
[14,]	"student"	"put"	"nation"	"discussion"	"serve"
[15,]	"space"	"officer"	"across"	"community"	"vietnam"
	Topic 21	Topic 22	Topic 23	Topic 24	Topic 25
[1,]	"funding"	"rep"	"county"	"house"	"national"
[2,]	"budget"	"members"	"residents"	"senate"	"security"
[3,]	"defense"	"congressman"	"fire"	"vote"	"public"
[4,]	"infrastructure"	"congress"	"valley"	"floor"	"secretary"

[5,]	"military"	"joined"	"area"	"colleagues"	"department"
[6,]	"critical"	"congressional"	"center"	"bill"	"social"
[7,]	"national"	"letter"	"local"	"republicans"	"homeland"
[8,]	"million"	"bipartisan"	"open"	"today"	"park"
[9,]	"billion"	"caucus"	"management"	"republican"	"nation's"
[10,]	"appropriations"	"colleagues"	"please"	"voted"	"border"
[11,]	"important"	"ryan"	"fort"	"pass"	"top"
[12,]	"needs"	"paul"	"evacuation"	"now"	"lands"
[13,]	"programs"	"sexual"	"st"	"resolution"	"alaska"
[14,]	"project"	"read"	"stay"	"spoke"	"western"
[15,]	"provides"	"fellow"	"areas"	"democrats"	"important"
	Topic 26	Topic 27	Topic 28	Topic 29	Topic 30
[1,]	"federal"	"jobs"	"school"	"live"	"communities"
[2,]	"government"	"businesses"	"high"	"watch"	"water"
[3,]	"congress"	"business"	"students"	"morning"	"local"
[4,]	"regulations"	"small"	"congressional"	"news"	"working"
[5,]	"agencies"	"economy"	"district"	"join"	"across"
[6,]	"spending"	"economic"	"service"	"tonight"	"state"
[7,]	"rule"	"create"	"academy"	"talk"	"food"
[8,]	"dollars"	"job"	"young"	"commerce"	"farmers"
[9,]	"wall"	"workers"	"learn"	"see"	"industry"
[10,]	"without"	"growth"	"year"	"discuss"	"agriculture"
[11,]	"real"	"help"	"competition"	"tune"	"central"
[12,]	"use"	"development"	"u.s"	"facebook"	"like"
[13,]	"debt"	"opportunities"	"interested"	"chamber"	"farm"
[14,]	"regulatory"	"grow"	"congratulations"	"tomorrow"	"rural"
[15,]	"money"	"local"	"app"	"page"	"great"
	Topic 31	Topic 32	Topic 33	Topic 34	Topic 35
[1,]	"make"	"act"	"tax"	"family"	"day"
[2,]	"can"	"bill"	"reform"	"great"	"women"
[3,]	"work"	"legislation"	"families"	"time"	"men"
[4,]	"sure"	"house"	"cuts"	"happy"	"today"
[5,]	"together"	"passed"	"working"	"everyone"	"every"
[6,]	"need"	"bipartisan"	"taxes"	"day"	"honor"
[7,]	"better"	"introduced"	"plan"	"many"	"thank"
[8,]	"making"	"financial"	"code"	"friends"	"nation"
[9,]	"ways"	"process"	"middle"	"today"	"lives"
[10,]	"come"	"h.r"	"class"	"thanks"	"country"
[11,]	"way"	"protection"	"pay"	"birthday"	"remember"
[12,]	"opportunity"	"bills"	"jobs"	"hope"	"lost"
[13,]	"bring"	"use"	"bill"	"best"	"sacrifice"
[14,]	"challenges"	"today"	"americans"	"celebrate"	"brave"
[15,]	"families"	"including"	"means"	"wish"	"service"

	Topic 36	Topic 37	Topic 38	Topic 39	Topic 40	Topic 41
[1,]	"health"	"years"	"work"	"american"	"free"	"first"
[2,]	"care"	"year"	"important"	"people"	"protect"	"violence"
[3,]	"affordable"	"one"	"hard"	"better"	"information"	"lives"
[4,]	"act"	"still"	"first"	"made"	"ensure"	"thoughts"
[5,]	"insurance"	"ago"	"step"	"deserve"	"internet"	"families"
[6,]	"healthcare"	"two"	"continue"	"time"	"public"	"victims"
[7,]	"repeal"	"time"	"done"	"back"	"must"	"prayers"
[8,]	"access"	"next"	"efforts"	"way"	"rules"	"gun"
[9,]	"obamacare"	"just"	"team"	"promise"	"companies"	"attack"
[10,]	"system"	"since"	"part"	"americans"	"control"	"steve"
[11,]	"costs"	"old"	"across"	"nothing"	"access"	"congress"
[12,]	"improve"	"today"	"go"	"put"	"clear"	"taken"
[13,]	"quality"	"past"	"hope"	"much"	"protections"	"shooting"
[14,]	"plans"	"nearly"	"can"	"deliver"	"personal"	"responders"
[15,]	"premiums"	"days"	"ensuring"	"promises"	"consumers"	"action"
	Topic 42	Topic 43	Topic 44	Topic 45	Topic 46	
[1,]	"community"	"america"	"support"	"forward"	"new"	
[2,]	"city"	"must"	"proud"	"look"	"u.s"	
[3,]	"texas"	"country"	"continue"	"working"	"air"	
[4,]	"dr"	"us"	"i'm"	"congress"	"force"	
[5,]	"university"	"world"	"fight"	"looking"	"york"	
[6,]	"center"	"nation"	"research"	"strong"	"mexico"	
[7,]	"de"	"stand"	"cancer"	"move"	"support"	
[8,]	"san"	"come"	"fighting"	"committed"	"ensure"	
[9,]	"la"	"always"	"stand"	"work"	"guard"	
[10,]	"el"	"together"	"keep"	"opportunity"	"facility"	
[11,]	"kansas"	"around"	"make"	"priorities"	"coast"	
[12,]	"king"	"across"	"risk"	"i'm"	"base"	
[13,]	"congressman"	"community"	"disease"	"many"	"state's"	
[14,]	"jr"	"threats"	"awareness"	"continuing"	"jersey"	
[15,]	"building"	"hate"	"advocates"	"serve"	"representatives"	
	Topic 47	Topic 48	Topic 49	Topic 50		
[1,]	"hall"	"can"	"veterans"	"just"		
[2,]	"questions"	"today"	"va"	"get"		
[3,]	"town"	"help"	"service"	"it's"		
[4,]	"join"	"get"	"care"	"one"		
[5,]	"event"	"open"	"ensure"	"going"		
[6,]	"staff"	"now"	"veteran"	"right"		
[7,]	"hope"	"find"	"affairs"	"people"		
[8,]	"please"	"plan"	"benefits"	"now"		
[9,]	"office"	"visit"	"department"	"time"		
[10,]	"may"	"sign"	"services"	"good"		

```
[11,] "hosting"      "family"      "medical"     "stop"
[12,] "constituents" "days"       "receive"     "think"
[13,] "call"        "less"        "access"      "back"
[14,] "hours"       "enrollment" "choice"      "another"
[15,] "county"      "deadline"    "served"      "i'm"
```

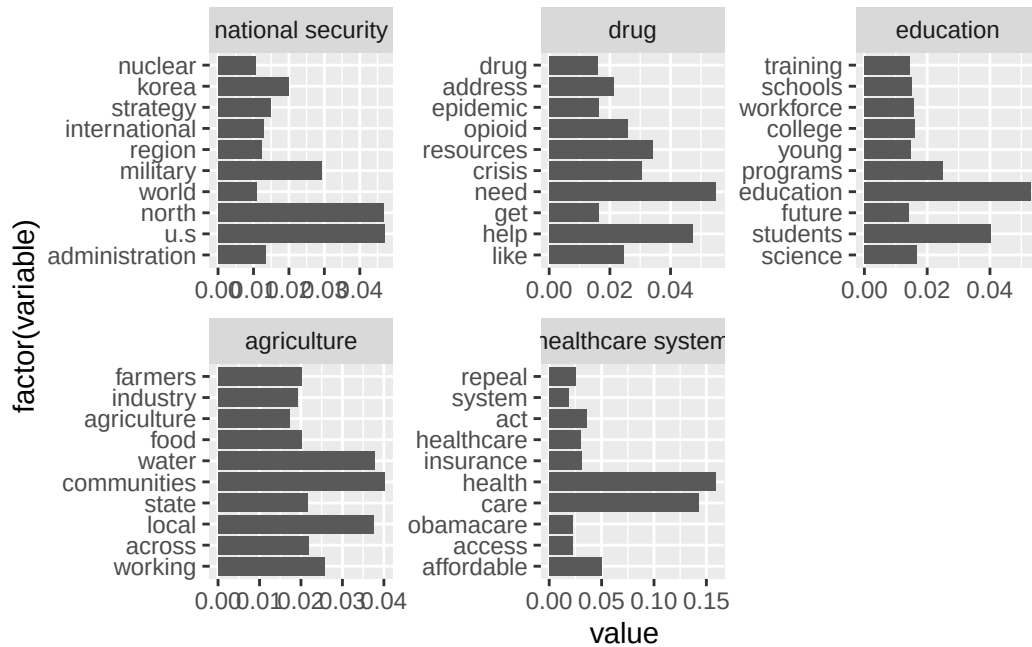
```
# get_terms(mylda, 15)[, c(3, 4, 16, 30, 36)]

# To get a more granular view, extract the probabilities
mylda_posterior <- topicmodels::posterior(object = mylda)
topic_distr_over_words <- mylda_posterior$terms
topic_distr_over_words_dt <- data.table(topic=1:K,
                                         topic_distr_over_words)
topic_distr_over_words_dt <- melt.data.table(topic_distr_over_words_dt, # If you use base R,
                                             id.vars = 'topic')

# Tidy/Dplyr way of extracting top 15 rows by group
top10per_topic <- topic_distr_over_words_dt %>%
  group_by(topic) %>%
  slice_max(order_by = value, n = 15)

# data.table way of extracting top 10 rows by group
topic_distr_over_words_dt <- topic_distr_over_words_dt[order(value,decreasing = T)]
top10per_topic <- topic_distr_over_words_dt[,head(.SD,10),by='topic']

topic_name <- c("national security", "drug", "education", "agriculture", "healthcare system")
# Plot probability over words for a few topics.
ggplot(top10per_topic[topic %in% c(3, 4, 16, 30, 36)],aes(y=factor(variable),x=value)) +
  geom_bar(stat = 'identity', position = 'dodge') +
  facet_wrap(~factor(topic,
                    levels = c(3, 4, 16, 30, 36),
                    labels = c("national security", "drug", "education", "agriculture", "healthcare system"),
                    ),
            scales = 'free')
```

15 topic 3 4 16 30 36
 topic 50 junk topic

5

```
# To validate labeling; identify documents with highest proportion on any given topic
doc_topic_proportions <- mylda_posterior$topics
# dim(doc_topic_proportions)
# head(doc_topic_proportions)
doc_topic_proportions_dt <- data.table(doc_id = mylda@documents,
                                       doc_topic_proportions)
# here I choose topic 3 and topic 36

colnames(doc_topic_proportions_dt)[2:ncol(doc_topic_proportions_dt)] <- paste0('Topic', colnames(doc_topic_proportions_dt)[2:ncol(doc_topic_proportions_dt)])

topic3doc <- doc_topic_proportions_dt[order(Topic3, decreasing = T)][1:3,]
topic36doc <- doc_topic_proportions_dt[order(Topic36, decreasing = T)][1:3,]

# Select rows from original data...
```

```
# national security
cat("Top 3 documents for topic 3 are:", topic3doc$doc_id)
```

Top 3 documents for topic 3 are: 4678 4201 2176

```
# fb_congress_data[doc_id %in% topic3doc$doc_id,]$message

# obamacare
cat("Top 3 documents for topic 36 are:", topic36doc$doc_id)
```

Top 3 documents for topic 36 are: 2405 4501 489

```
# fb_congress_data[doc_id %in% topic36doc$doc_id,]$message
```

topic 3 topic 36 national security healthcare system.

6.

```
set.seed(1)
K <- 3
mylda_3 <- LDA(x = dtm,
               k = K,
               method="Gibbs",
               control=list(iter = 1000,
                           seed = 1,
                           verbose = 0))

# Inspect top 10 from each topics
get_terms(mylda_3, 10)
```

	Topic 1	Topic 2	Topic 3
[1,]	"national"	"president"	"health"
[2,]	"u.s"	"house"	"care"
[3,]	"great"	"bill"	"today"
[4,]	"new"	"tax"	"veterans"
[5,]	"help"	"american"	"families"

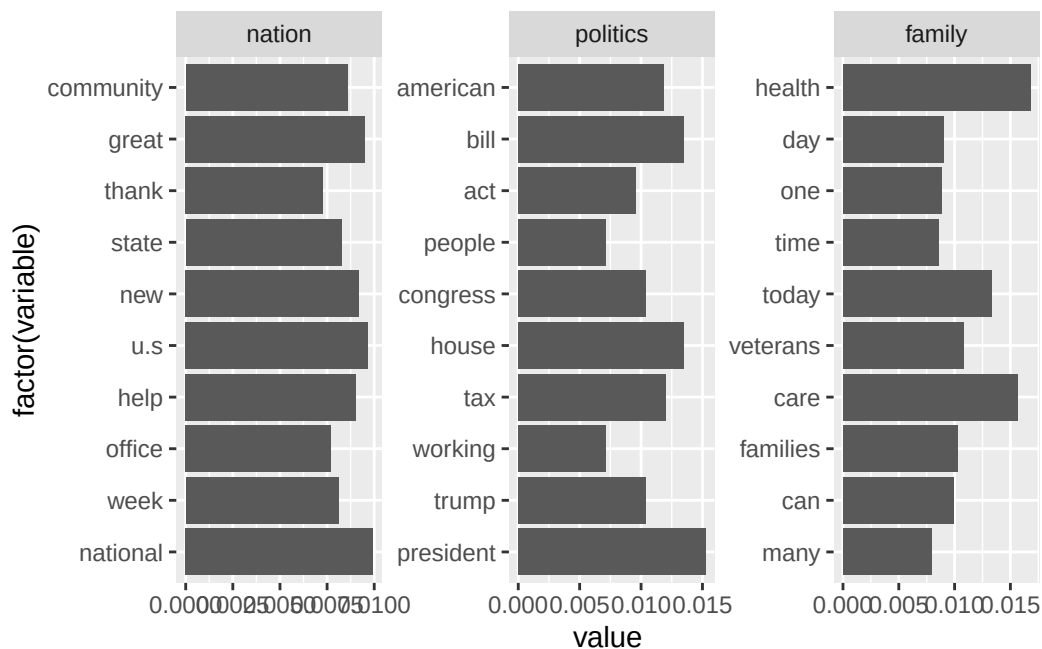
```
[6,] "community" "trump"      "can"
[7,] "state"     "congress"  "day"
[8,] "week"      "act"       "one"
[9,] "office"    "working"   "time"
[10,] "thank"    "people"    "many"
```

```
# To get a more granular view, extract the probabilities
mylda_3_posterior <- topicmodels::posterior(object = mylda_3)
topic_distr_over_words <- mylda_3_posterior$terms
topic_distr_over_words_dt <- data.table(topic=1:K,
                                         topic_distr_over_words)
topic_distr_over_words_dt <- melt.data.table(topic_distr_over_words_dt, # If you use base R,
                                             id.vars = 'topic')

# Tidy/Dplyr way of extracting top 15 rows by group
top10per_topic <- topic_distr_over_words_dt %>%
  group_by(topic) %>%
  slice_max(order_by = value, n = 15)

# data.table way of extracting top 10 rows by group
topic_distr_over_words_dt <- topic_distr_over_words_dt[order(value,decreasing = T)]
top10per_topic <- topic_distr_over_words_dt[,head(.SD,10),by='topic']

# Plot probability over words for a few topics.
ggplot(top10per_topic[topic %in% 1:3],aes(y=factor(variable),x=value)) +
  geom_bar(stat = 'identity', position = 'dodge') +
  facet_wrap(~factor(topic,
                     labels = c("nation", "politics", "family")
                   ),
            scales = 'free')
```



```
# To validate labeling; identify documents with highest proportion on any given topic
doc_topic_proportions <- mylda_3_posterior$topics
dim(doc_topic_proportions)
```

```
[1] 5748    3
```

```
head(doc_topic_proportions)
```

```
      1      2      3
1 0.2979798 0.4191919 0.2828283
2 0.3777778 0.2944444 0.3277778
3 0.2348485 0.4393939 0.3257576
4 0.4278607 0.2935323 0.2786070
5 0.2635659 0.4031008 0.3333333
6 0.2666667 0.4196078 0.3137255
```

```
doc_topic_proportions_dt <- data.table(doc_id = mylda_3@documents,
                                       doc_topic_proportions)
colnames(doc_topic_proportions_dt)[2:ncol(doc_topic_proportions_dt)] <- paste0('Topic', colnames(doc_topic_proportions_dt)[2:ncol(doc_topic_proportions_dt)])
topic2doc <- doc_topic_proportions_dt[order(Topic2, decreasing = T)][1:3,]
```

```
topic3doc <- doc_topic_proportions_dt[order(Topic3,decreasing = T)][1:3,]
cat("Top 3 documents for topic 2 are:", topic2doc$doc_id)
```

Top 3 documents for topic 2 are: 1744 2443 1899

```
cat("Top 3 documents for topic 3 are:", topic3doc$doc_id)
```

Top 3 documents for topic 3 are: 4108 290 857

```
# Select rows from original data...
# fb_congress_data[doc_id==1744,]$message
# fb_congress_data[doc_id==4108,]$message
```

K=50 K=3 K=3

7.

```
posterior_results <- posterior(mylda)
theta <- posterior_results$topics
topic_prevalence <- colMeans(theta)

cat("The most prevalence is topic", which.max(topic_prevalence))
```

The most prevalence is topic 33

```
labeled_topics <- c(1:length(topic_prevalence))

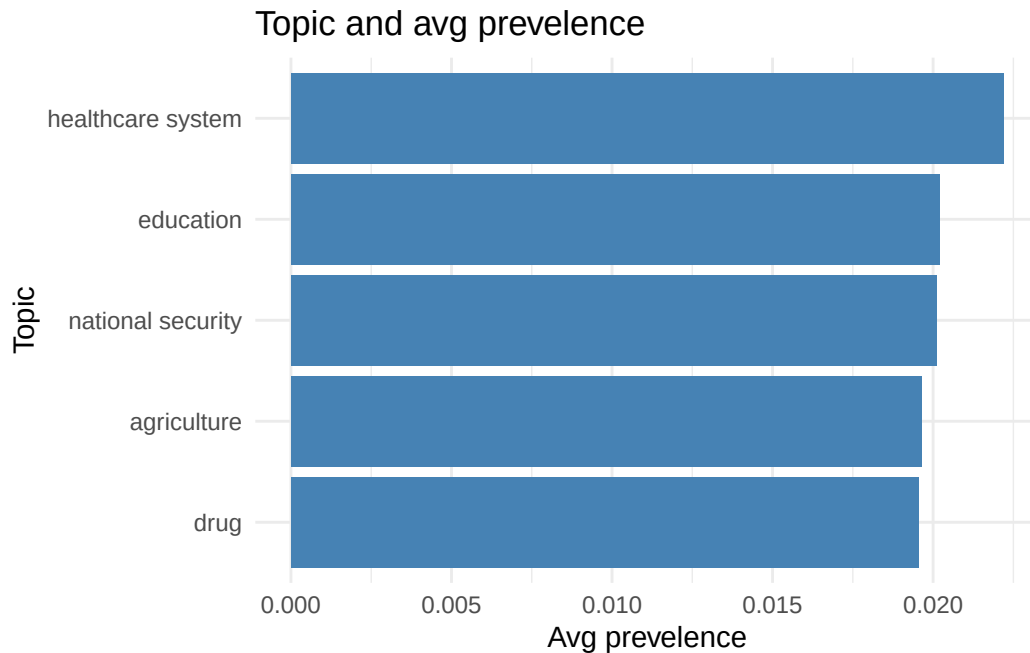
plot_data <- data.frame(
  topic = paste0("Topic ", labeled_topics),
  prevalence = topic_prevalence[labeled_topics]
)
```



```

    title = "Topic and avg prevalence"
  ) +
  theme_minimal()

```



topic 33

```

party_raw <- docvars(dtm, "party")

K <- ncol(theta)
colnames(theta) <- paste0("Topic ", 1:K)

theta_df <- as.data.frame(theta) %>%
  mutate(party = party_raw)

theta_long <- theta_df %>%
  pivot_longer(cols = starts_with("Topic"),
               names_to = "topic",
               values_to = "proportion")

labeled_topics <- paste0("Topic ", c(3, 4, 16, 30, 36))

theta_long_labeled <- theta_long %>% filter(topic %in% labeled_topics)

```

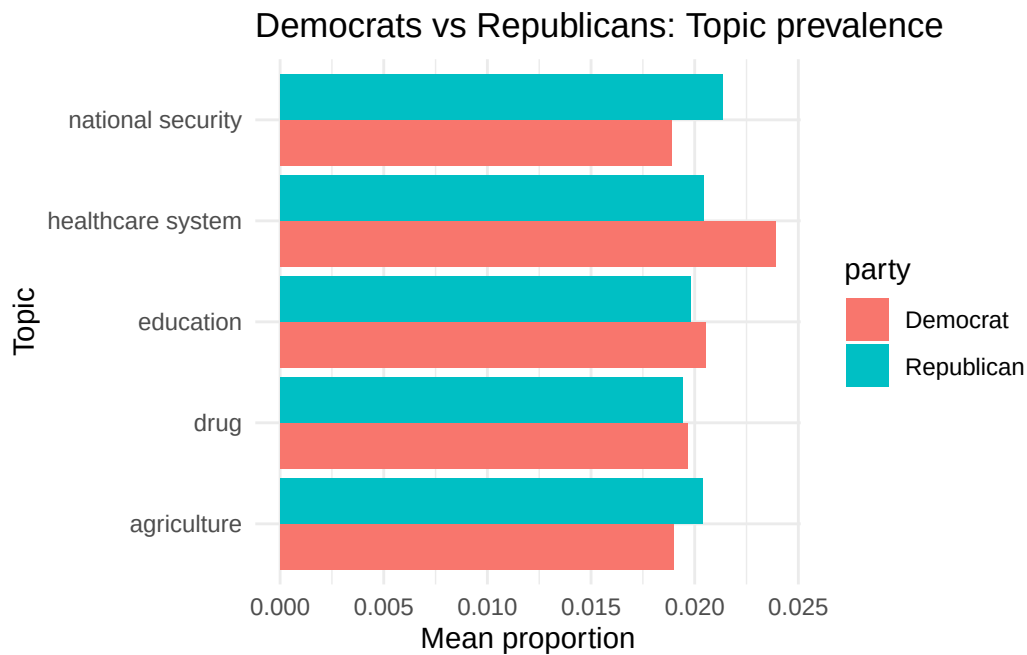
```

mapping <- c(
  "Topic 3" = "national security",
  "Topic 4" = "drug",
  "Topic 16" = "education",
  "Topic 30" = "agriculture",
  "Topic 36" = "healthcare system"
)

theta_long_labeled$topic_label <- mapping[theta_long_labeled$topic]

ggplot(theta_long_labeled,
  aes(x = topic_label, y = proportion, fill = party)) +
  stat_summary(fun = mean, geom = "bar", position = "dodge") +
  labs(x = "Topic", y = "Mean proportion",
    title = "Democrats vs Republicans: Topic prevalence") +
  theme_minimal() +
  coord_flip()

```



national security, healthcare system, agriculture , democrat republican


```

# glm

glm_topic30 <- glm(`Topic 30` ~ party, family = "quasibinomial", data = theta_df)
# summary(glm_topic30)$coefficients
# broom::tidy(glm_topic30, exponentiate = TRUE)

glm_results <- lapply(labeled_topics, function(tnm) {
  frm <- as.formula(paste0("`", tnm, "` ~ party"))
  fit <- glm(frm, family = "quasibinomial", data = theta_df)
  broom::tidy(fit, exponentiate = TRUE) |> mutate(topic = tnm)
}) |> bind_rows()

glm_wide <- glm_results |>
  select(topic, term, estimate) |>
  pivot_wider(names_from = term, values_from = estimate) |>
  mutate(topic_label = mapping[topic]) |>
  relocate(topic_label, .after = topic)

glm_wide |> kable()

```

topic	topic_label	(Intercept)	partyRepublican
Topic 3	national security	0.0192342	1.1351985
Topic 4	drug	0.0200616	0.9858144
Topic 16	education	0.0209708	0.9644434
Topic 30	agriculture	0.0193341	1.0751342
Topic 36	healthcare system	0.0245023	0.8502475

3 odds ratio 1.135 13.5% 36 odds ratio 0.850
15% 4 16 30 odds ratio 1

```

# t test
res <- t.test(theta_df[["Topic 3"]] ~ theta_df$party)
# res$p.value

t_results <- lapply(labeled_topics, function(tnm){
  res <- t.test(theta_df[[tnm]] ~ theta_df$party)
  data.frame(
    topic = tnm,
    p_value = res$p.value,
    mean_in_Democrat = res$estimate[1],

```

```

    mean_in_Republican = res$estimate[2]
  )
}
) |>
  bind_rows()
rownames(t_results) <- NULL

t_results <- t_results |>
  mutate(topic_label = mapping[topic]) |>
  relocate(topic_label, .after = topic)

t_results |> kable()

```

topic	topic_label	p_value	mean_in_Democrat	mean_in_Republican
Topic 3	national security	0.0000331	0.0188712	0.0213681
Topic 4	drug	0.5487106	0.0196671	0.0193935
Topic 16	education	0.1754252	0.0205400	0.0198242
Topic 30	agriculture	0.0024109	0.0189674	0.0203634
Topic 36	healthcare system	0.0000001	0.0239163	0.0204079

t-test topic 3 topic 30 topic 36 topic 4 topic 16

8.

```

set.seed(1984)
row_ids <- 1:nrow(dtm)
train_ids <- sample(x = row_ids, size = 0.8*length(row_ids), replace = F)
test_ids <- row_ids[!row_ids %in% train_ids]
train_dt <- dtm[train_ids,]
test_dt <- dtm[-train_ids,]

ks <- c(3, 10, 25, 50, 75, 100, 200)
perp <- c()
for(i in 1:length(ks)){
  current_tm <- LDA(x = train_dt,
                    k = ks[i],
                    method="Gibbs",

```

```

        control=list(iter = 500,
                      seed = 1,
                      verbose = 0))

# Compute perplexity
perp[i] <- perplexity(object = current_tm,
                     newdata = as.simple_triplet_matrix(test_dt)) # For some reason, perp.
print(perp)
}

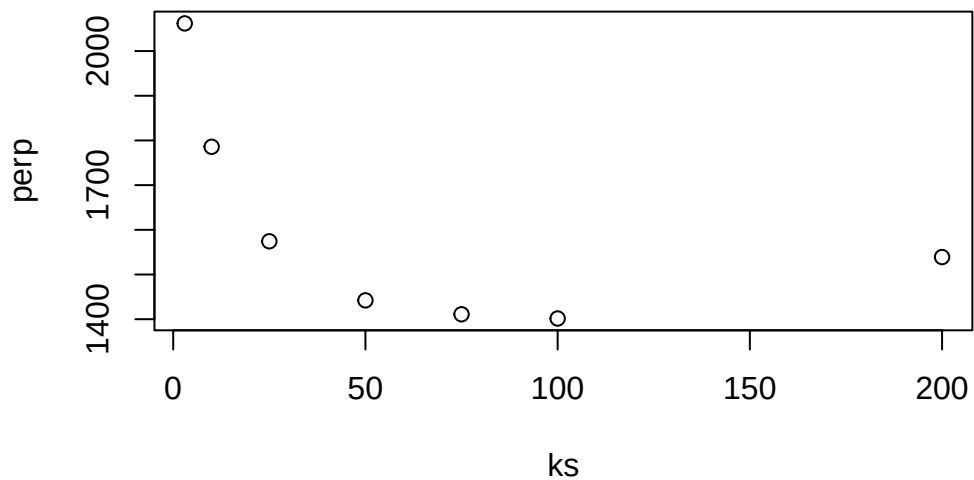
```

```

[1] 2061.894
[1] 2061.894 1785.798
[1] 2061.894 1785.798 1574.329
[1] 2061.894 1785.798 1574.329 1442.176
[1] 2061.894 1785.798 1574.329 1442.176 1411.016
[1] 2061.894 1785.798 1574.329 1442.176 1411.016 1401.556
[1] 2061.894 1785.798 1574.329 1442.176 1411.016 1401.556 1539.033

```

```
plot(x=ks,y=perp)
```



According to the elbow method, $K = 50$ is the best K perplexity,
 50 75 100 interpretable

Part 2:

1.

```
fb_congress_data <- fread("~/Documents/ML/Labs/Lab5/fb-congress-data3.csv")
```

```
posts_corpus <- corpus(x = fb_congress_data,
                       text_field = "message",
                       meta = list("screen_name", "party"),
                       docid_field = 'doc_id')

# print(posts_corpus)

# Tokenize & clean from particular types of words
mytokens <- tokens(x = posts_corpus,
                  remove_punct = TRUE,
                  remove_numbers = TRUE,
                  remove_symbols = TRUE,
                  remove_url = TRUE,
                  padding = FALSE)

# print(mytokens)

# Make tokens lowercase
mytokens <- tokens_tolower(x = mytokens)

# How does it look now?
# mytokens[1:3]

mysentences <- sapply(mytokens, function(x) paste(x, collapse = " "))
# mysentences[1:3]
```

2.

```
set.seed(1984)
w2v <- word2vec(x = mysentences, # Your data
               type = "cbow",     # Model type (the one we talked about in lecture)
               window = 15,       # Context defined in terms of +-5 words.
               dim = 50,          # Dimensionality of embeddings
               iter = 50,         # Estimation iterations (higher means more time)
```

```

hs = FALSE,          # Setting to FALSE here --> "negative sampling
negative = 15)       # Number of negative samples (how many fake sent

# w2v_association <- word2vec(x = mysentences,          # Your data
#                             type = "skip-gram",      # Model type (the one we talked about
#                             window = 15,           # Context defined in terms of +-5 words.
#                             dim = 50,              # Dimensionality of embeddings
#                             iter = 50,             # Estimation iterations (higher means more t
#                             hs = FALSE,            # Setting to FALSE here --> "negative sampli
#                             negative = 15)         # Number of negative samples (how many fake s

```

3.

```

selected_words <- c("russia", "tax", "trump")
w2v_predict <- predict(w2v, selected_words, type = "nearest", top_n = 10)

w2v_predict

```

```

$russia
  term1      term2 similarity rank
1  russia    russian  0.8752062    1
2  russia    inquiry  0.8503195    2
3  russia investigation  0.8231674    3
4  russia    collusion  0.7965655    4
5  russia    connections  0.7959847    5
6  russia    interference  0.7895319    6
7  russia      putin  0.7775518    7
8  russia    investigated  0.7757707    8
9  russia    election  0.7748210    9
10 russia    interfere  0.7739654   10

```

```

$tax
  term1      term2 similarity rank
1   tax      taxes  0.8025434    1
2   tax    standard  0.7329622    2
3   tax    windfall  0.7041653    3
4   tax   draconian  0.6981813    4
5   tax   taxreform  0.6928915    5
6   tax   trillions  0.6820951    6
7   tax        adds  0.6675219    7

```

8	tax	simplifies	0.6651972	8
9	tax	immigration	0.6550444	9
10	tax	overseas	0.6527149	10

```
$trump
  term1    term2 similarity rank
1 trump trump's  0.9279581     1
2 trump  obama  0.8854365     2
3 trump   vice  0.8304774     3
4 trump obama's  0.8045565     4
5 trump  reagan  0.7619424     5
6 trump kennedy  0.7575178     6
7 trump  elect  0.7252138     7
8 trump   tone  0.7123275     8
9 trump incoming 0.7024204     9
10 trump  hiding 0.7013357    10
```

skip-gram

4.

```
embedding <- as.matrix(w2v)
```

```
embedding[rownames(embedding)=="women",]
```

```
[1]  0.253035694  0.363205194 -1.741279960 -0.599990964 -0.148379251
[6] -1.560149908  0.904341877 -0.177421868 -0.118451945 -0.979910672
[11] -0.857758820  0.524356842  0.167056367  0.624618948  0.723112941
[16] -0.476479530 -0.720045984  1.279627800 -1.136861801  0.404365778
[21] -1.103777289  2.073764801 -0.509807944 -0.838766396  0.061711837
[26]  0.005485578 -0.100816458  1.819810748  0.070811793 -1.371308088
[31]  1.060047865 -0.319657713  1.076126099  0.414122373  1.358442187
[36] -0.981719196  1.102052927  0.800468504 -1.713262439 -1.436105132
[41]  2.365748882 -0.264750451 -0.721847355  0.383461475  1.137141585
[46]  0.583636642  1.147403121 -1.726913929 -0.021738909  0.403602690
```

```
kingtowoman <- embedding["king", ] - embedding["man", ] + embedding["woman", ]
sims <- word2vec_similarity(embedding, matrix(kingtowoman, nrow = 1))
```

```
head(sort(sims[, 1], decreasing = TRUE), 20)
```

king	luther	martin	legacy
1.0301168	0.9250198	0.8774292	0.8454438
holocaust	woman	enduring	achievement
0.8381134	0.8238519	0.7843010	0.7822881
mlkday	shameful	commit	harper
0.7810862	0.7746442	0.7682997	0.7672731
determination	blackhistorymonth	spirit	bipartisanship
0.7658074	0.7594079	0.7566742	0.7558194
abolitionist	pulse	strive	learn
0.7552724	0.7535951	0.7480589	0.7464680

```
pretrained <- readRDS("~/Documents/ML/Labs/Lab5/glove6B200d.rds")
dim(pretrained)
```

```
[1] 400000    200
```

```
# embedding_pretrained <- as.matrix(pretrained) # pretrained is "matrix" "array"
# pretrained[rownames(pretrained)=="king",]
kingtowoman <- pretrained["king", ] - pretrained["man", ] + pretrained["woman", ]
sims <- word2vec_similarity(pretrained, matrix(kingtowoman, nrow = 1))
head(sort(sims[, 1], decreasing = TRUE), 20)
```

king	queen	princess	prince	throne	emperor	royal	daughter
0.4630444	0.4272459	0.4001166	0.3994429	0.3877838	0.3765438	0.3732692	0.3721567
monarch	kingdom	crown	mother	her	marriage	wife	elizabeth
0.3719335	0.3668355	0.3652451	0.3617875	0.3608811	0.3567054	0.3565992	0.3557918
woman	duchess	adulyadej	duke				
0.3553641	0.3549088	0.3543496	0.3536174				

```
dist_vec <- function(a,b) sqrt(sum((a-b)^2))
dist_sta_man <- dist_vec(pretrained["statistician", ], pretrained["man", ])
dist_sta_woman <- dist_vec(pretrained["statistician", ], pretrained["woman", ])
occupationalbias <- dist_sta_man - dist_sta_woman
occupationalbias
```

```
[1] -0.3029102
```

```
occupational bias -0.303  statistician embedding  man woman  occupational bias
```

5.

```
corpus_dem <- corpus_subset(posts_corpus, party == "Democrat")
corpus_rep <- corpus_subset(posts_corpus, party == "Republican")

mytokens_dem <- tokens(x = corpus_dem,
  remove_punct = TRUE,
  remove_numbers = TRUE,
  remove_symbols = TRUE,
  remove_url = TRUE,
  padding = FALSE)

mytokens_rep <- tokens(x = corpus_rep,
  remove_punct = TRUE,
  remove_numbers = TRUE,
  remove_symbols = TRUE,
  remove_url = TRUE,
  padding = FALSE)

# Make tokens lowercase
mytokens_dem <- tokens_tolower(x = mytokens_dem)
mytokens_rep <- tokens_tolower(x = mytokens_rep)

mysentences_dem <- sapply(mytokens_dem, function(x) paste(x, collapse = " "))
mysentences_rep <- sapply(mytokens_rep, function(x) paste(x, collapse = " "))
```

```
set.seed(1984)
w2v_dem <- word2vec(x = mysentences_dem,
  type = "cbow",
  window = 15,
  dim = 50,
  iter = 50,
  hs = FALSE,
  negative = 15)
# Your data
# Model type (the one we talked about in lecture)
# Context defined in terms of +-5 words.
# Dimensionality of embeddings
# Estimation iterations (higher means more time)
# Setting to FALSE here --> "negative sampling"
# Number of negative samples (how many fake sentences)

w2v_rep <- word2vec(x = mysentences_rep,
  type = "cbow",
  window = 15,
  dim = 50,
  iter = 50,
  hs = FALSE)
# Your data
# Model type (the one we talked about in lecture)
# Context defined in terms of +-5 words.
# Dimensionality of embeddings
# Estimation iterations (higher means more time)
# Setting to FALSE here --> "negative sampling"
```



```
negative = 15) # Number of negative samples (how many fake sen
```

```
predict(w2v_dem, c("immigrants", "abortion"), type = "nearest", top_n = 10)
```

```
$immigrants
      term1      term2 similarity rank
1 immigrants dreamers  0.7226679    1
2 immigrants refugees  0.7076225    2
3 immigrants granted   0.7063723    3
4 immigrants solidarity 0.6949468    4
5 immigrants immigrant 0.6939174    5
6 immigrants sisters   0.6859382    6
7 immigrants values    0.6859117    7
8 immigrants individuals 0.6839132    8
9 immigrants circuit    0.6679727    9
10 immigrants refugee   0.6674896   10
```

```
$abortion
      term1      term2 similarity rank
1 abortion restrictions 0.7092915    1
2 abortion reproductive 0.6921242    2
3 abortion focused      0.6833248    3
4 abortion minority     0.6809095    4
5 abortion threatens    0.6727515    5
6 abortion largest      0.6708782    6
7 abortion nutrition    0.6564741    7
8 abortion dod          0.6544619    8
9 abortion 30am         0.6538859    9
10 abortion pregnant    0.6497677   10
```

```
predict(w2v_rep, c("immigrants", "abortion"), type = "nearest", top_n = 10)
```

```
$immigrants
      term1      term2 similarity rank
1 immigrants deported  0.8363445    1
2 immigrants aliens    0.8300606    2
3 immigrants dangerous 0.8210338    3
4 immigrants illegal   0.7939292    4
5 immigrants sanctuary 0.7638013    5
6 immigrants violent   0.7548273    6
```

7	immigrants	offenders	0.7516261	7
8	immigrants	criminal	0.7451375	8
9	immigrants	results	0.7292443	9
10	immigrants	cooperate	0.7263024	10

\$abortion

	term1	term2	similarity	rank
1	abortion	parenthood	0.7822995	1
2	abortion	unborn	0.7774661	2
3	abortion	planned	0.7434731	3
4	abortion	x	0.7279820	4
5	abortion	pregnancy	0.7249730	5
6	abortion	babies	0.6844142	6
7	abortion	original	0.6812647	7
8	abortion	cosponsor	0.6709932	8
9	abortion	capable	0.6607584	9
10	abortion	parents	0.6537328	10

immigrants

immigrants

abortion

beliefs

6

```
positive <- fread("~/Documents/ML/Labs/Lab5/positive.txt")[[1]]
negative <- fread("~/Documents/ML/Labs/Lab5/negative.txt")[[1]]
selected_words <- c("russia", "tax", "trump")
```

```
embeddings_dem <- as.matrix(w2v_dem)
pos_in_dem <- intersect(positive, rownames(embeddings_dem))
neg_in_dem <- intersect(negative, rownames(embeddings_dem))
pos_matrix_dem <- embeddings_dem[pos_in_dem, , drop = FALSE]
neg_matrix_dem <- embeddings_dem[neg_in_dem, , drop = FALSE]
pos_vec_dem <- colMeans(pos_matrix_dem)
neg_vec_dem <- colMeans(neg_matrix_dem)

sentiment_vec_dem <- neg_vec_dem - pos_vec_dem
sims_dem <- word2vec_similarity(embeddings_dem, sentiment_vec_dem)
similarity_table_dem <- data.frame(
  word = rownames(embeddings_dem),
  sim = as.numeric(sims_dem[, 1])
)
```

```
sims_dem_filtered <- similarity_table_dem |>
  filter(!(word %in% negative)) |>
  arrange(desc(sim)) |>
  mutate(rank = row_number())
```

```
embeddings_rep <- as.matrix(w2v_rep)
pos_in_rep <- intersect(positive, rownames(embeddings_rep))
neg_in_rep <- intersect(negative, rownames(embeddings_rep))
pos_matrix_rep <- embeddings_rep[pos_in_rep, , drop = FALSE]
neg_matrix_rep <- embeddings_rep[neg_in_rep, , drop = FALSE]
pos_vec_rep <- colMeans(pos_matrix_rep)
neg_vec_rep <- colMeans(neg_matrix_rep)

sentiment_vec_rep <- neg_vec_rep - pos_vec_rep
sims_rep <- word2vec_similarity(embeddings_rep, sentiment_vec_rep)
similarity_table_rep <- data.frame(
  word = rownames(embeddings_rep),
  sim = as.numeric(sims_rep[, 1])
)
sims_rep_filtered <- similarity_table_rep |>
  filter(!(word %in% negative)) |>
  arrange(desc(sim)) |>
  mutate(rank = row_number())
```

```
sims_dem_filtered |> head(10)
```

	word	sim	rank
1	highly	0.3742807	1
2	result	0.3604322	2
3	cbo	0.3557619	3
4	withdraw	0.3474114	4
5	threatened	0.3433241	5
6	muslimban	0.3430340	6
7	abbott	0.3428643	7
8	halt	0.3376548	8
9	raises	0.3362890	9
10	marketplaces	0.3359981	10

```
sims_rep_filtered |> head(10)
```

	word	sim	rank
--	------	-----	------

1	spread	0.3832878	1
2	certain	0.3826789	2
3	crimes	0.3764794	3
4	wildfire	0.3738432	4
5	terrorists	0.3683356	5
6	ignoring	0.3682224	6
7	deficit	0.3673807	7
8	probation	0.3591972	8
9	canceled	0.3574313	9
10	massive	0.3563566	10

```
rank_dem <- sims_dem_filtered |> filter(word %in% selected_words)

rank_rep <- sims_rep_filtered |> filter(word %in% selected_words)

rank_dem |>
  kable(caption = "Word Rank for Democrats")
```

Table 3: Word Rank for Democrats

word	sim	rank
trump	0.2678148	198
russia	0.2036654	636
tax	0.1958383	699

```
rank_rep |>
  kable(caption = "Word Rank for Republicans")
```

Table 4: Word Rank for Republicans

word	sim	rank
russia	0.3230826	54
tax	0.0427675	1728
trump	0.0000000	2122

trump, russia, gun tax trump similarity trump trump similarity trump

```

fit_and_rank <- function(mysentences, positive_words = positive, negative_words = negative, type = "pos", window = 1, dim = 100, iter = 1, hs = 1, negative = negative) {
  w2v <- word2vec(x = mysentences, type = type, window = window, dim = dim,
                  iter = iter, hs = hs, negative = negative)

  emb <- as.matrix(w2v)
  pos_in <- intersect(positive_words, rownames(emb))
  neg_in <- intersect(negative_words, rownames(emb))
  pos_vec <- colMeans(emb[pos_in, , drop = FALSE])
  neg_vec <- colMeans(emb[neg_in, , drop = FALSE])
  sentiment_vec <- neg_vec - pos_vec
  sims <- word2vec_similarity(emb, sentiment_vec)
  similarity_table <- data.frame(
    word = rownames(emb),
    sim = as.numeric(sims[, 1])
  )
}

sims_filtered <- similarity_table |>
  filter(!(word %in% negative_words)) |>
  arrange(desc(sim)) |>
  mutate(rank = row_number())
}

bootstrap_by_party <- function(mysentences, positive_words = positive, negative_words = negative, seed = 1234) {
  set.seed(seed)
  map_dfr(seq_len(n_boot), function(bi) {
    idx <- sample.int(length(mysentences), replace = TRUE)
    sent_boot <- mysentences[idx]
    fit_and_rank(sent_boot, positive_words = positive, negative_words = negative) |>
      filter(word %in% selected_words) |>
      mutate(iter = bi)
  })
}

boot_dem <- bootstrap_by_party(mysentences_dem) |>
  mutate(party = "Democrat")
boot_rep <- bootstrap_by_party(mysentences_rep) |>
  mutate(party = "Republican")

boot_all <- bind_rows(boot_dem, boot_rep)

# saveRDS(boot_all, file = "boot_all.rds")

```

```

boot_summary <- boot_all |>
  group_by(party, word) |>
  summarize(n_iter = n(),
            sim_mean = mean(sim, na.rm = TRUE),
            sim_low = quantile(sim, 0.025, na.rm = TRUE),
            sim_high = quantile(sim, 0.975, na.rm = TRUE),
            rank_mean = mean(rank, na.rm = TRUE),
            .groups = "drop")

boot_sim_wide <- boot_summary |>
  select(party, word, sim_mean, sim_low, sim_high) |>
  pivot_wider(names_from = party, values_from = c(sim_mean, sim_low, sim_high))

boot_summary |> kable()

```

party	word	n_iter	sim_mean	sim_low	sim_high	rank_mean
Democrat	russia	20	0.1911417	0.1329699	0.2405574	525.05
Democrat	tax	20	0.1508933	0.0534016	0.2177232	869.20
Democrat	trump	20	0.2290203	0.2040539	0.2509196	229.15
Republican	russia	20	0.2708425	0.2242510	0.3318392	110.40
Republican	tax	20	0.0381105	0.0000000	0.1231368	1829.05
Republican	trump	20	0.0186684	0.0000000	0.1091795	2085.10

```
boot_sim_wide |> kable()
```

word	sim_mean_Democrat	sim_mean_Republican	sim_low_Democrat	sim_low_Republican	sim_high_Democrat	sim_high_Republican
russia	0.1911417	0.2708425	0.1329699	0.224251	0.2405574	0.3318392
tax	0.1508933	0.0381105	0.0534016	0.000000	0.2177232	0.1231368
trump	0.2290203	0.0186684	0.2040539	0.000000	0.2509196	0.1091795

similarity

russia 0 russia